

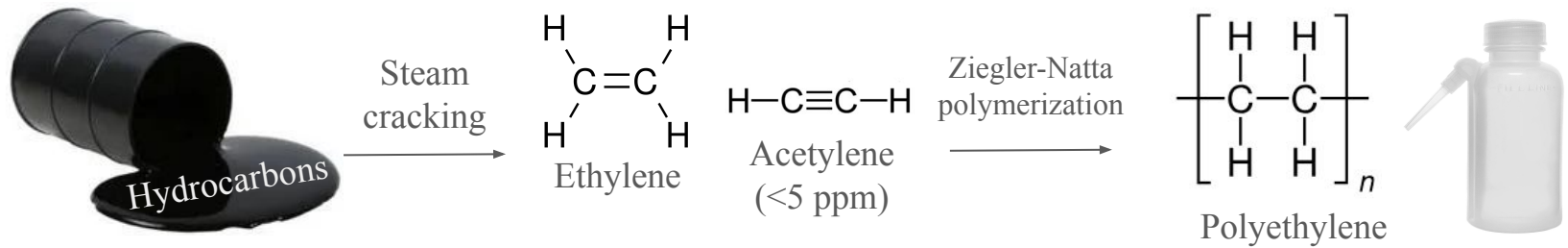
Semi-hydrogenation of Acetylene using Metal Organic Frameworks (MOF) Catalysts

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Ch232

References

1. Gonçalves, L. P. L.; Wang, J.; Vinati, S.; Barborini, E.; Wei, X.-K.; Heggen, M.; Franco, M.; Sousa, J. P. S.; Petrovykh, D. Y.; Soares, O. S. G. P.; Kovnir, K.; Akola, J.; Kolen'ko, Y. V. Combined Experimental and Theoretical Study of Acetylene Semi-Hydrogenation over Pd/Al₂O₃. *International Journal of Hydrogen Energy* **2020**, 45 (2), 1283–1296. <https://doi.org/10.1016/j.ijhydene.2019.04.086>.
2. Iliescu, A.; Oppenheim, J. J.; Sun, C.; Dincă, M. Conceptual and Practical Aspects of Metal–Organic Frameworks for Solid–Gas Reactions. *Chem. Rev.* **2023**, 123 (9), 6197–6232. <https://doi.org/10.1021/acs.chemrev.2c00537>.
3. Tejeda-Serrano, M.; Mon, M.; Ross, B.; Gonell, F.; Ferrando-Soria, J.; Corma, A.; Leyva-Pérez, A.; Armentano, D.; Pardo, E. Isolated Fe(III)–O Sites Catalyze the Hydrogenation of Acetylene in Ethylene Flows under Front-End Industrial Conditions. *J. Am. Chem. Soc.* **2018**, 140 (28), 8827–8832. <https://doi.org/10.1021/jacs.8b04669>.
4. Gao, F.; Yan, R.; Shu, Y.; Cao, Q.; Zhang, L. Strategies for the Application of Metal–Organic Frameworks in Catalytic Reactions. *RSC Adv.* **2022**, 12 (16), 10114–10125. <https://doi.org/10.1039/D2RA01175A>.
5. Ji, S.; Chen, Y.; Zhao, S.; Chen, W.; Shi, L.; Wang, Y.; Dong, J.; Li, Z.; Li, F.; Chen, C.; Peng, Q.; Li, J.; Wang, D.; Li, Y. Atomically Dispersed Ruthenium Species Inside Metal–Organic Frameworks: Combining the High Activity of Atomic Sites and the Molecular Sieving Effect of MOFs. *Angew Chem Int Ed* **2019**, 58 (13), 4271–4275. <https://doi.org/10.1002/anie.201814182>.
6. Mahajan, S.; Lahtinen, M. Recent Progress in Metal–Organic Frameworks (MOFs) for CO₂ Capture at Different Pressures. *Journal of Environmental Chemical Engineering* **2022**, 10 (6), 108930. <https://doi.org/10.1016/j.jece.2022.108930>.

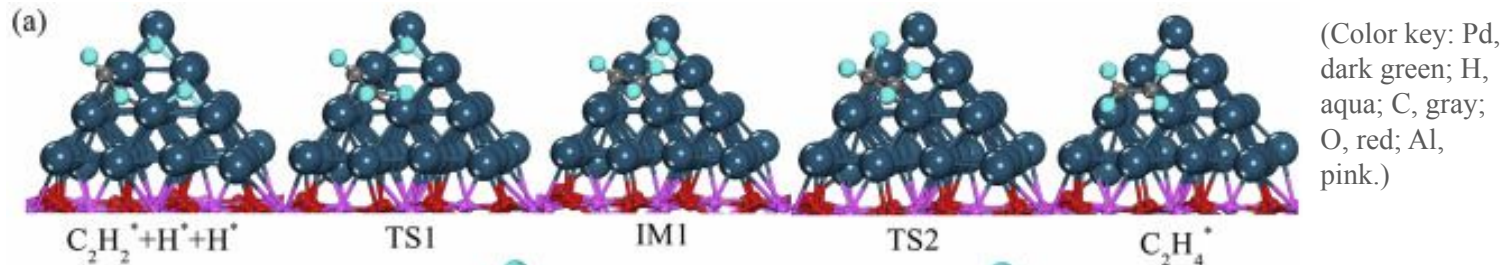
Want:



But:

- Ethylene feedstock produced from steam cracking hydrocarbons usually contains 0.5% to 2.0% acetylene
- Significant presence of acetylene poisons Ziegler-Natta catalyst and explosive acetylides could be synthesized instead
- Must purify feedstock through the semi-hydrogenation of acetylene to ethylene
- The catalyst must also be very selective so that ethylene is not converted into ethane

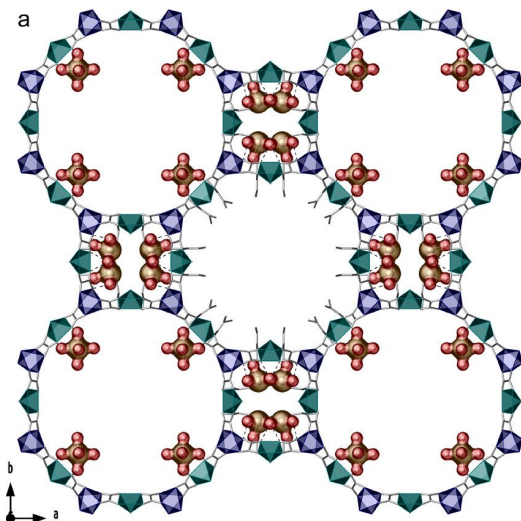
Pd-based catalysts:
(ex: Pd/Al₂O₃)



- Although Pd-based catalysts have high activities for acetylene conversion, they also exhibit low selectivity and limited stability → need a better catalyst!
- **Metal-Organic Frameworks (MOFs):** lattices composed of organic linkers and inorganic secondary building units
 - Highly porous, have large surface areas, contain dispersed metallic active sites
 - Modifiable active sites, linkers, secondary building units → can be extremely selective chemically and physically

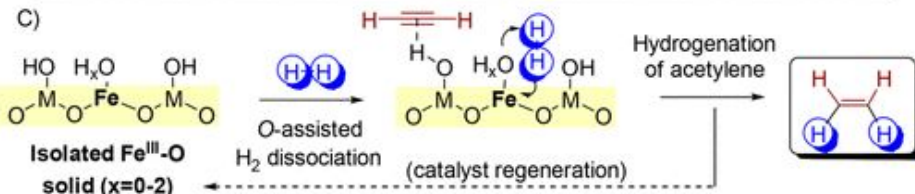
Iron Oxide-based MOF (Tejada-Serrano et al.)

$[\text{Fe}^{\text{III}}(\text{H}_2\text{O})_6][\text{Fe}_2^{\text{III}}(\mu\text{-O})_2(\text{H}_2\text{O})_6]_{1/2}\{\text{Ni}^{\text{II}}_4[\text{Cu}^{\text{II}}_2(\text{Me}_3\text{mpba})_2]_3\}\cdot 72\text{H}_2\text{O}$, where $\text{Me}_3\text{mpba}^{4-}$ is a *N,N'*-2,4,6-trimethyl-1,3-phenylenebis(oxamate) ligand



Crystal structure of iron oxide-based MOF determined by XRD

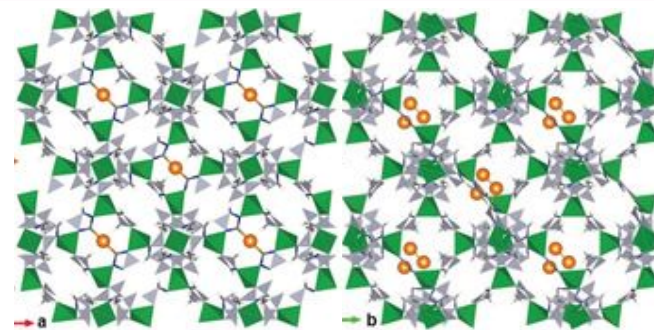
(Key: Copper(II) cations, cyan polyhedra; nickel(II) cations, blue polyhedra; ligands, gray sticks; iron(III) cations, gold; water molecules, red spheres; dotted lines, hydrogen bonds between the network and the iron units)



- Reduced the amount of acetylene from 1.2% to <10 ppm
- H_2 dissociates into the active $\text{Fe}^{\text{III}}\text{-O}$ site, transfers to a nearby acetylene, which coordinates and exchanges H atoms with neighboring hydroxyl groups

Ru-based MOF (Ji et al.)

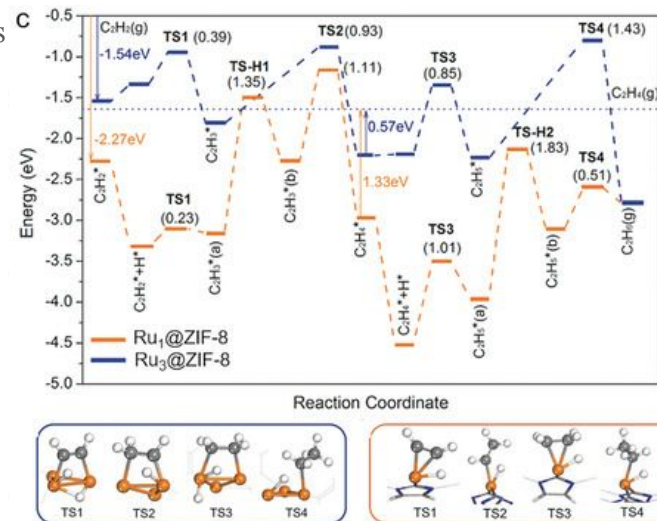
$\text{Ru}_1@\text{ZIF-8}$ and $\text{Ru}_3@\text{ZIF-8}$, where ZIFs are zeolitic imidazolate frameworks (Zn^{2+} connected by imidazolate ($\text{C}_3\text{H}_3\text{N}_2^-$) linkers)



$\text{Ru}_1@\text{ZIF-8}$ (left) and $\text{Ru}_3@\text{ZIF-8}$ (right)

(Key: Ru, yellow; Zn(II) cation, green; imidazolate linkers, gray)

- Acetylene bonds to Ru atoms
- Ru_1 : H migrates from ZIF-8 to Ru
- Ru_3 : H from another Ru
- Ru_3 50 times more active
- Ru_3 more selective (84% vs. 51%) (molecular sieving effect)



Limitations:

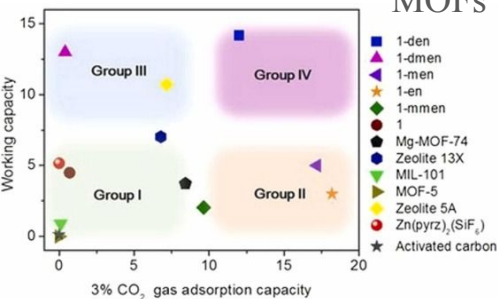
- Catalyst deactivation over time: degradation of active sites, formation of deposits that block pores
 - Ex: For iron oxide-based MOF, increased ethane formation over time → Fe–O active site lost its selectivity as the reaction progressed
- Uniformly distributing active metal sites
- Long-term thermal and chemical stability under harsh reaction conditions

Advantages:

- MOFs are not prone to demetalation and leaching (leading to an improvement in lifetime)
- Do not require solvents (resulting in less material and energy costs and less chemical waste)
- Can be easily recycled and reused
- Typically require cheaper materials than Pd (very expensive!)

Interesting Application:

- Direct air capture (DAC): removes CO_2 directly from the atmosphere
- Low concentration of CO_2 in ambient air (~417 ppm) requires materials with high selectivity, capacity, and regenerability → MOFs are a promising solution!



Recent developments in MOF families like MOF-74 and amine-functionalized variants have shown significant potential for use in DAC systems

